

VehicleArena: A Realistic Urban Environment for Multi-Agent Driving

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Abstract

Real-world embodied agents often pursue independent objectives within a shared physical environment, where their actions can alter the conditions faced by others. Existing benchmarks, however, typically assume shared goals or explicitly prescribed interaction protocols, leaving such emergent physical coupling underexplored. We introduce **VehicleArena**, a 3D urban-driving benchmark for studying independently operating agents in a dynamic shared world. In VehicleArena, LLM-controlled agents must fulfill evolving passenger requests while navigating complex traffic, and each agent’s driving decisions can reshape traffic flow, delays, risks, and subsequent observations for surrounding agents. The benchmark provides 112 evaluation tasks—80 single-agent and 32 multi-agent. Across nine evaluated models, the highest arrival rates reach only 65.0% on single-agent tasks and 65.6% on multi-agent tasks, while strong passenger-request or cabin scores do not reliably translate into successful trip completion. Moreover, in matched multi-agent runs, every evaluated driving policy reduces the arrival rate of surrounding vehicles relative to the simulator’s native traffic controller, revealing measurable externalities beyond the ego vehicle itself.

Github: <https://github.com/OpenMOSS/VehicleArena>

Dataset: <https://huggingface.co/datasets/OpenMOSS-Team/VehicleArena>

1 Introduction

Language and vision-language agents are increasingly being studied in embodied driving settings, yet existing work has largely progressed along two separate tracks. Closed-loop driving benchmarks focus on on-road decision making, evaluating an ego vehicle’s multimodal perception, planning, and instruction-following capabilities [5, 12, 16]. In parallel, intelligent cockpit environments emphasize in-cabin interaction and device control [15]. Although both settings capture important agent capabilities, they are typically evaluated in isolation: passenger intent and cabin interaction are separated from the physical driving process. This makes it difficult to study how an agent should respond to evolving passenger requests while continuously adapting to a changing road environment.

To bridge this gap, we introduce VehicleArena, a high-fidelity 3D urban-driving benchmark that unifies passenger interaction, cockpit control, and physical driving within a single closed loop. VehicleArena integrates passenger-oriented Personal Agents with LLM-controlled Driving Agents. Personal Agents issue

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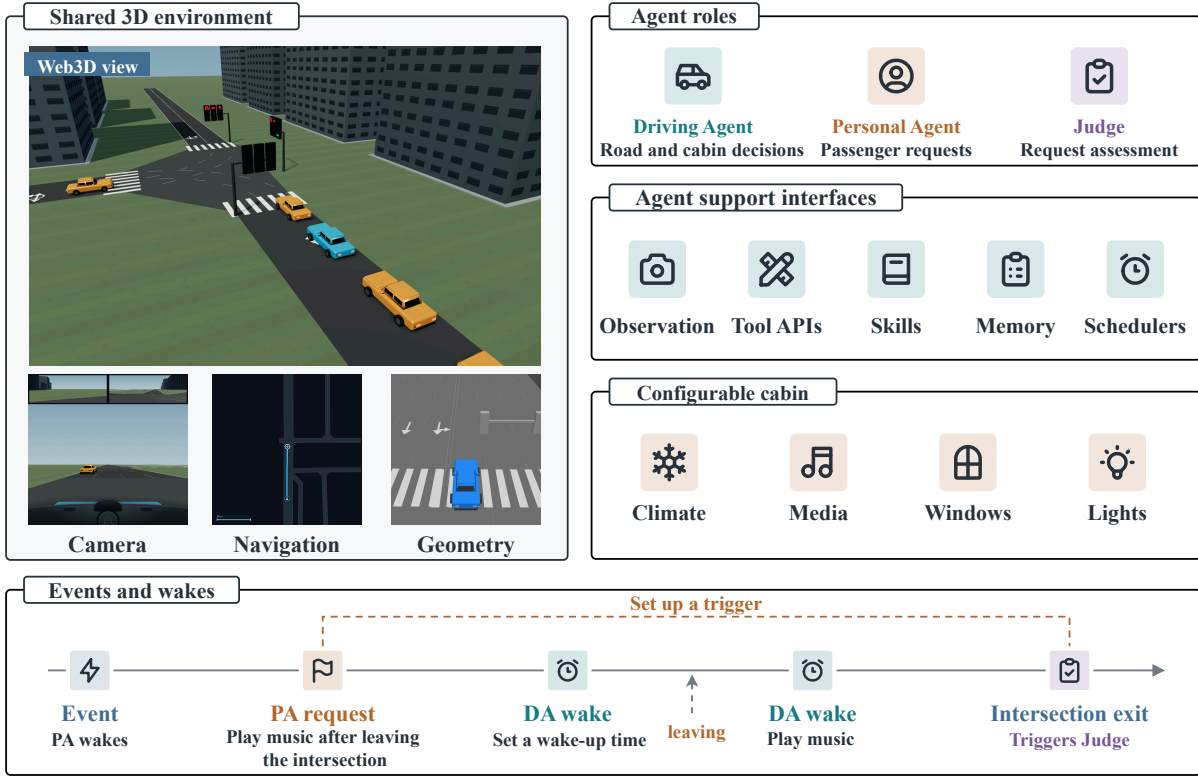


Figure 1 VehicleArena architecture. Left: the shared 3D world and observation examples. Right: agent roles, support interfaces, and configurable cabin capabilities. Bottom: PA asks DA to play music after leaving an intersection and registers a verification trigger; DA sets a wake-up time to carry out the request, while Judge checks are triggered independently of DA wakes. The timeline is illustrative and not to scale. The main image is a native Web3D capture from a SUMO-only reference rollout; the smaller images are separate observation examples.

requests involving routes, driving preferences, and in-vehicle devices, while Driving Agents must interpret and execute these requests under dynamic traffic conditions. As a result, passenger intent, cabin operations, route decisions, and vehicle motion evolve together along a shared timeline.

This unified design naturally extends to multi-vehicle settings. VehicleArena supports multiple LLM-controlled vehicles operating concurrently in the same urban environment, with each vehicle serving its own passenger and pursuing its own objectives. Unlike conventional multi-agent benchmarks, where agent relationships are typically defined through explicit collaboration, competition, or communication structures [7, 13, 18, 19], agents in VehicleArena need not share goals, plans, or interaction protocols. Instead, they influence one another through the physical consequences of their actions in a shared road environment.

A route change, merge, yield, sudden stop, or hesitation by one vehicle can immediately alter the observations, risks, and feasible actions of surrounding agents. We refer to this setting as independent-objective physical coexistence: independently motivated agents become coupled not through a predefined social structure, but through persistent physical interaction in a shared world. This setting allows us to ask a central question: can LLM agents remain reliable and safe when their behavior must account for both changing user demands and the unpredictable consequences of other independently acting agents?

To support research in this setting, VehicleArena provides 100 development tasks and a held-out evaluation set of 112 tasks. The evaluation set includes 80 single-agent tasks that assess integrated driving and passenger-request execution, and 32 MultiLLM interaction tasks that probe multi-agent physical coexistence. We evaluate Driving Agents using metrics spanning trip completion, driving quality, passenger-request

satisfaction, cabin correctness, traffic externalities, and inference cost.

Our contributions are summarized as follows:

- We develop a high-fidelity 3D agentic-driving environment that unifies multimodal driving, passenger-side objectives, intelligent cockpit interaction, and dynamic urban traffic within a single closed loop.
- We extend the environment to multi-vehicle scenarios and formulate independent-objective physical coexistence, where agents with independent objectives interact through the physical consequences of their actions in a shared world.
- We construct a benchmark with 112 held-out evaluation tasks covering both integrated single-agent capabilities and multi-agent physical coexistence, together with metrics for driving, passenger-request execution, traffic impact, and inference cost.

2 Related Work

2.1 Multi-Agent LLM Systems

Multi-agent LLM systems often organize agents around a shared objective. CAMEL uses role-playing dialogue, while ChatDev and MetaGPT assign specialized roles and workflows for collaborative software development [6, 7, 11]. PARTNR extends this paradigm to embodied human–robot collaboration [2]. Other benchmarks introduce private or competitive objectives: SOTOPIA studies agents with individual social goals, while MultiAgentBench and BattleAgentBench cover cooperation and competition under predefined interaction structures [13, 18, 19]. Across these settings, agent relationships are largely specified by the task, whether as teammates, opponents, or negotiation partners.

VehicleArena instead studies physical coexistence among agents with independent objectives. Agents serve different users and need not share goals, plans, or communication protocols. Their interactions emerge through a shared physical environment: a lane change, yield, or sudden stop by one vehicle changes the risks and feasible actions of nearby agents. VehicleArena therefore focuses on whether independently operating agents remain reliable when their coupling arises from persistent physical interaction rather than a predefined social structure.

2.2 LLM-Based Autonomous Driving

Recent language- and vision-language-based driving methods primarily study scene understanding and ego-vehicle decision making. SGDrive structures driving knowledge for VLM-based planning [8], while NAVSIM and Pseudo-Simulation enable scalable evaluation through non-reactive rollouts [1, 3]. DriveBench further evaluates the visual grounding and robustness of VLM-based driving reasoning [14]. These settings expose important perception and planning failures, but do not execute decisions in a persistent, reactive traffic world.

Closed-loop simulators such as CARLA, SMARTS, MetaDrive, and SUMO provide interactive traffic dynamics [4, 9, 10, 17], and systems including LMDrive, DriveArena, and LimSim++ integrate multimodal LLMs into such environments [5, 12, 16]. However, they remain largely centered on an ego vehicle following a predefined navigation objective or instruction. In parallel, VehicleWorld studies LLM interaction with in-vehicle devices, but separates cockpit interaction from physical driving [15].

VehicleArena unifies these settings in a persistent environment where passenger requests, cabin state, driving decisions, and surrounding traffic evolve together. It further supports multiple independently controlled vehicles, connecting integrated passenger–driver interaction with the physical multi-agent coexistence described above.

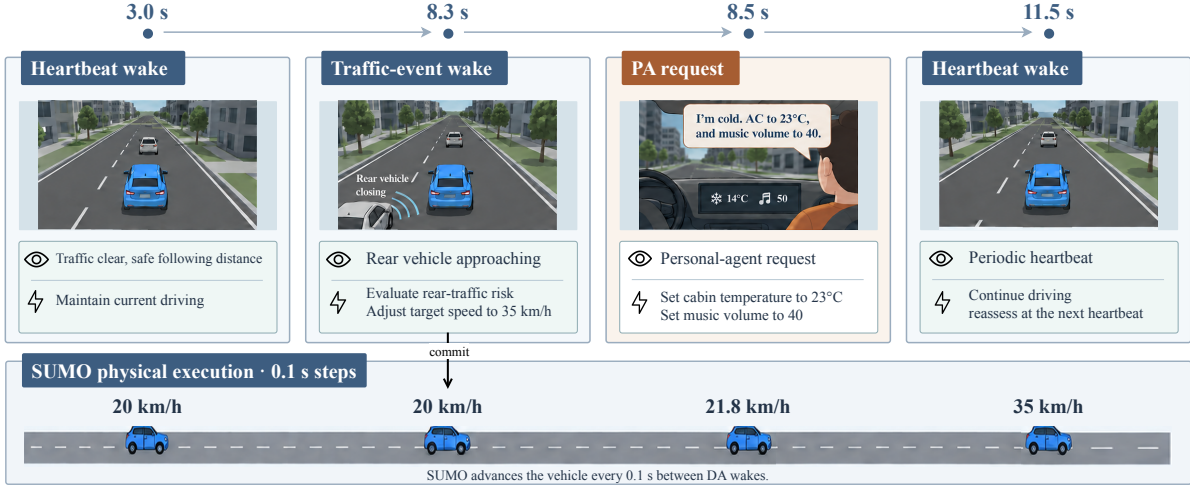


Figure 2 A recorded episode illustrating interleaved driving and passenger service. The upper panels show three DA wakes and a passenger request, together with the relevant observations and actions. Following a rear-traffic warning, DA raises its target speed to 35 km/h; it later handles a cabin request without replacing that motion target. The lower road shows the corresponding speeds realized by SUMO, which advances at 0.1-second steps between DA wakes. Time gaps are compressed and intervening wakes are omitted.

3 VehicleArena Environment

VehicleArena brings passenger service and physical driving into a shared urban world. We first introduce the system components, and then describe the harnesses that connect agent decisions, physical execution, and request verification.

3.1 Overview

Figure 1 illustrates a shared 3D environment in which traffic, cabin state, and environmental events evolve along a single simulation timeline. SUMO advances vehicle motion and traffic interactions, while synchronized rendering exposes the resulting world state to the agents. Vehicles can differ in physical capabilities and installed equipment; extensible cabin modules and a rule engine define available device operations and operating constraints. Each participating vehicle can be controlled either by an LLM-based driver or by native SUMO logic, allowing selected background vehicles to become independent agents without changing the surrounding environment. Vehicle configuration is detailed in Appendix E.

Three agent roles operate over the same evolving episode. A Personal Agent (PA) expresses passenger needs, a Driving Agent (DA) determines how to serve them while driving, and a Judge evaluates whether execution satisfies the request. All three refer to the same world state and simulation clock, but have distinct responsibilities and information access: PA cannot directly actuate the vehicle, while Judge observes and evaluates execution without controlling it. In multi-vehicle scenarios, each DA serves its own passenger and destination. Their objectives remain independent, while their actions become physically interdependent through the shared road environment. Appendix H presents their prompt templates; Appendix J lists the tools available to each role.

3.2 Driving-Agent Harness

Agent loop. DA interacts with the world through repeated observe–act sessions, as illustrated in Figure 2. At each wake, the harness combines current observations and passenger updates with retained task context. DA may inspect the situation, invoke tools, and use their feedback over multiple turns before ending the session. Recent interaction history and a persistent Todo board preserve unfinished work across wakes.

DA is not invoked at every physics step. In addition to wakes triggered by startup or relevant events, it controls a periodic heartbeat and may schedule a future wake, making when to reason part of its policy. Between wakes, the world continues to evolve under the currently active controls. When multiple DAs wake at the same simulation time, they observe the same committed road state and run in parallel; their buffered motion commands are committed before the next physics step.

Tool discovery and skills. The harness exposes capabilities progressively so that the same agent loop can operate across differently equipped vehicles. Core driving tools are immediately available, while additional capabilities can be discovered by inspecting installed modules and loading their corresponding interfaces. Optional skills provide procedural guidance for multi-step operations without prescribing a driving strategy. DA can therefore discover and invoke the capabilities required by a passenger request during an ongoing episode. The skill catalog and complete guide contents are provided in Appendix I.

Driving action space. DA controls the vehicle through persistent target-speed commands, ordinary or emergency braking, adjacent-lane changes, and junction maneuvers such as turning or continuing straight. Route planning provides navigation guidance rather than automatic route following; DA remains responsible for selecting the required maneuvers and signaling. These commands specify intended control rather than instantaneous state changes. The simulator realizes them under the vehicle’s physical constraints and surrounding traffic, so braking takes time and a requested lane change may be constrained by nearby road users. Cabin tools similarly modify persistent device state rather than merely producing textual responses.

3.3 Personal-Agent and Judge Harness

PA wakes on passenger-visible events or randomized timers. It observes outstanding requests and currently feasible trigger options, and may issue, revise, or cancel both immediate and delayed requests. Once a request is accepted, its desired outcomes and acceptance criteria are fixed. DA receives the passenger update but independently determines how to execute the request and when to wake again.

For delayed requests, the simulator monitors the trigger selected by PA and schedules Judge checks independently of DA wakes. Judge evaluates the fixed criteria against recorded interactions and physical execution history, incorporating earlier checks when a requirement must hold over time. A request remains pending until completion or its final verification point, while cases that cannot be reliably validated remain unscored. The complete loop is shown in Algorithm 1 in Appendix A.3; scoring details are provided in Section 4.3.

4 Benchmark

The benchmark uses the system in Section 3 to evaluate whether agents can complete their own trips and passenger tasks while sharing the road with others. We describe how tasks are constructed, how matched SUMO runs provide a comparison and time budget, and which outcomes are reported.

4.1 Task Suite and Construction

The suite contains 100 training and development tasks and 112 held-out evaluation tasks. The evaluation set comprises 80 BASIC tasks with one DA-controlled vehicle and 32 MULTILLM tasks with several independently configured DAs. Appendices A.2 and F summarize the task split and supported cities.

Tasks are constructed under human expert supervision, with experts participating in the design of traffic situations, vehicle trips, and passenger goals. The resulting tasks cover intersections, merges, narrow roads, and pedestrian crossings, requiring agents to complete their trips and respond to passenger requests.

4.2 Evaluation Protocol

Oracle baseline and time budget. We treat SUMO’s native driving policy as a driving oracle. Before evaluating a DA, we run each scene with SUMO controlling the ego vehicle to establish reference driving behavior,

traffic outcomes, and a task deadline. The baseline and evaluated runs use exactly the same environment and surrounding-agent configurations; only the ego vehicle’s controller changes. In MULTILLM, the other LLM-controlled vehicles use Qwen3.8-27B with distinct personality prompts and active PAs, all held fixed across the two runs. This oracle baseline provides a reference for assessing both the DA’s own driving behavior and its effects on other road users, including additional collisions, non-arrivals, and delays. To set the deadline, we record the last arrival time T_{cal} among required SUMO-controlled vehicles, excluding persistent obstacles and without waiting for LLM-controlled peers to arrive. With T_{event} denoting the last scheduled environmental event, we set

$$T_{\text{limit}} = \max(T_{\text{cal}}, T_{\text{event}}) + \Delta. \quad (1)$$

The margin Δ is normally 10 seconds. The resulting deadline is fixed throughout evaluation.

4.3 Evaluation Metrics

We report six complementary outcomes: trip completion and driving quality for the ego vehicle, passenger-request fulfillment and cabin compliance for service, effects on background traffic relative to the SUMO baseline, and model-token use for efficiency. These outcomes remain separate rather than being combined into one score. Table 9 in Appendix A.4 gives their definitions and reporting units; Appendices A.5 and D detail passenger grading and driving-rule deductions.

5 Experiments

5.1 Experimental Setup

We evaluate nine DA models on 112 held-out tasks: 80 BASIC and 32 MULTILLM scenarios. Each model is evaluated three times per task. In BASIC, only the ego vehicle uses the evaluated DA; other traffic participants retain native simulation control. In MULTILLM, the ego DA shares the road with three other DA-controlled vehicles. Each peer uses Qwen3.8-27B with a distinct personality prompt and an active PA. Their model, prompt, and PA assignments remain the same across model comparisons, although their actions respond to the evolving traffic.

The main comparison uses the same task IDs and scoring rules for each model. Traffic-impact analyses pair evaluated runs with SUMO references by task and required vehicle IDs. Model specifications are listed in Appendix G.

5.2 Main Results

Table 1 compares ego-vehicle outcomes in the two task groups; effects on surrounding vehicles are analyzed separately below.

Basic shows a wide spread in ego-vehicle arrival. Qwen3.8-Max and DeepSeek-V4.1-Flash reach 65.0% and 62.5% arrival, respectively, while MIMO-V2.6-Pro has the highest Request score at 90.0. GPT-5.6-Sol has the highest Cabin score at 90.2 but reaches only 20.0% arrival. This pattern is consistent with insufficient progress before the task deadline despite strong cabin performance.

Multi-Agent results show a similar separation between arrival and other scores. Qwen3.8-Max reaches 65.6% arrival, followed by MIMO-V2.6-Pro and GLM-5.3-Flash at 62.5% each. GPT-5.6-Sol records the highest Drive and Cabin scores, 87.2 and 94.0, but reaches only 9.4% arrival. Across all tasks it also has the lowest mean ego speed (11.4 km/h) and no recorded collisions (Table 2), consistent with limited progress before the task deadline.

Across both task groups, the order of models changes across the four metrics: strong driving, passenger-request, or cabin scores do not consistently coincide with successful arrival. Reporting these outcomes separately exposes the gap between completing a trip and satisfying other parts of the task.

Table 1 Main evaluation by task group. Basic has 80 tasks per model and Multi-Agent has 32. The best value in each metric column is bold; the second-best value is underlined, including ties.

Model	Basic (80 tasks)				Multi-Agent (32 tasks)			
	Arr. (%)	Drive	Req.	Cabin	Arr. (%)	Drive	Req.	Cabin
DeepSeek-V4.1-Flash	<u>62.5</u>	85.0	<u>88.5</u>	<u>76.6</u>	56.3	81.7	82.1	<u>73.9</u>
MIMO-V2.6-Pro	40.0	79.5	90.0	<u>57.5</u>	<u>62.5</u>	80.0	<u>88.0</u>	<u>66.7</u>
Kimi-K3	51.3	78.9	84.9	41.5	<u>40.6</u>	71.4	<u>76.4</u>	49.2
GLM-5.3-Flash	45.0	78.6	85.7	55.6	<u>62.5</u>	80.9	85.7	63.1
Qwen3.8-Max	65.0	<u>83.1</u>	86.7	72.0	65.6	<u>82.4</u>	88.2	65.4
Qwen3.8-27B	36.3	<u>79.6</u>	83.1	47.4	40.6	<u>74.0</u>	86.7	54.5
GPT-5.6-Sol	20.0	82.5	80.8	90.2	9.4	87.2	78.1	94.0
Qwen3-VL-8B	17.5	39.7	39.0	11.6	18.8	44.5	40.0	7.0
Qwen3-VL-32B	21.3	49.3	60.2	34.3	25.0	53.3	58.2	31.8

5.3 Analysis

We inspect the ego vehicle’s recorded decisions to characterize how each model spends inference and tool calls, and how it drives. We report Basic and Multi-Agent tasks separately for the quantitative comparisons. For the plots, we combine the four metrics in Table 1: ego arrival, Drive, Request, and Cabin. We average the latter three scores, then weight that average by the ego vehicle’s arrival rate. This completion-weighted index captures trip completion and task quality on one outcome axis for comparing planning effort; Appendix A.7 gives the exact formula.

Planning efficiency varies across models. We compare models within each task group by outcome and interaction cost. A short failed run is not good planning merely because it uses little interaction. Figure 3 shows wake frequency and tool calls per wake; together these determine tool calls per task. Qwen3.8-Max illustrates the favorable pattern: on Basic it reaches nearly the same outcome as DeepSeek-V4.1-Flash with fewer wakes and fewer tool calls per task, and it remains strong on Multi-Agent. GLM-5.3-Flash and MIMO-V2.6-Pro have similar Multi-Agent outcomes, but GLM uses fewer wakes and fewer calls within each wake. GLM-5.3-Flash and Kimi-K3 have similar wake frequencies, tool calls per wake, and outcomes on Basic. Their interaction counts remain similar on Multi-Agent, but GLM achieves a much stronger outcome than Kimi. This contrast shows that interaction volume alone does not explain their different outcomes in shared traffic.

GPT-5.6-Sol illustrates how planning activity can become unproductive: repeated wakes with few tool calls per wake accumulate without strong outcomes. Wake count also depends on episode duration and events, and call count says nothing about whether an observation or action was timely.

Overall, Qwen3.8-Max pairs strong outcomes with economical interaction, while frequent shallow wakes do not ensure success. GLM and Kimi show that similar interaction counts can yield different outcomes. Appendix B extends this analysis with skill, tool, and token statistics; Section B.2 compares token use with the same outcome index.

Speed and collisions characterize physical driving. Table 2 pools all tasks for each model. Qwen3.8-Max has the highest mean speed, few ego collisions, and the strongest arrival in both task groups. MIMO-V2.6-Pro and GPT-5.6-Sol have no ego collisions, but MIMO moves faster and arrives more often. GLM-5.3-Flash and Kimi-K3 have similar mean speeds, yet GLM reaches substantially more Multi-Agent destinations. Their Basic arrival rates are closer, which the pooled speed cannot reveal. These contrasts show that neither speed nor collision count alone captures effective driving.

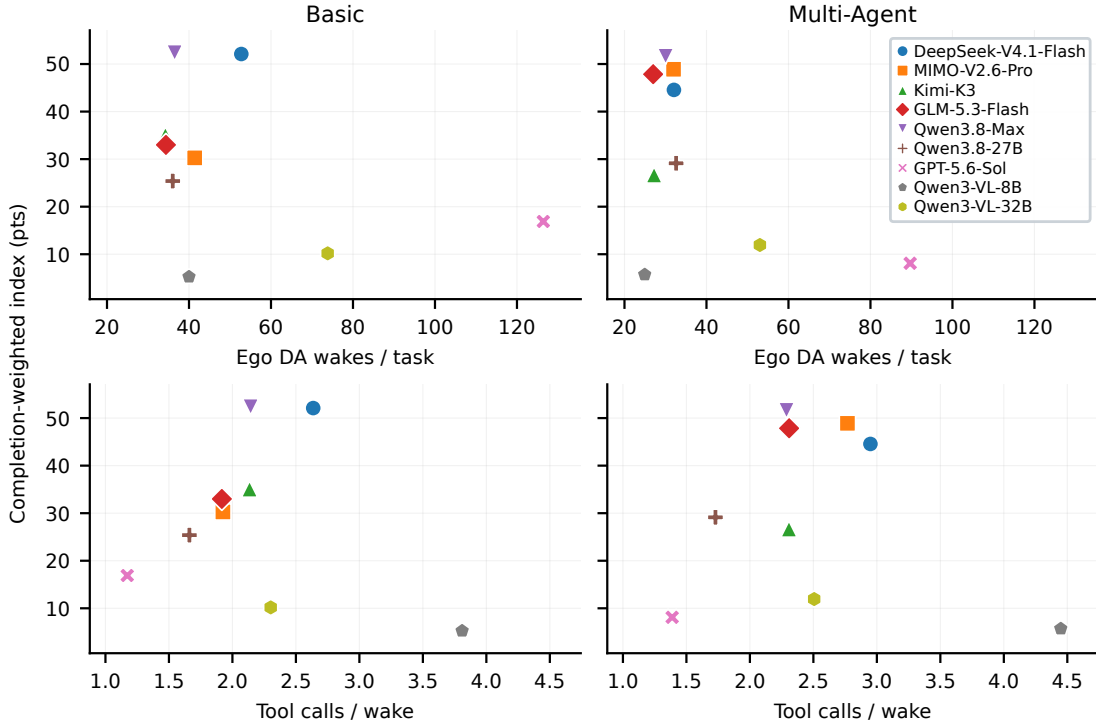


Figure 3 Planning activity versus completion-weighted outcome for Basic (left) and Multi-Agent (right): ego DA wakes per task (top) and tool calls per wake (bottom). Each marker denotes a model; Appendix A.7 defines the outcome index.

Table 2 Ego vehicle speed and collisions across all 112 tasks. Mean speed includes stopped time; collisions count ego-involved events.

Model	Speed (km/h)	Collisions
DeepSeek-V4.1-Flash	19.0	4
MIMO-V2.6-Pro	16.8	0
Kimi-K3	18.6	4
GLM-5.3-Flash	18.5	3
Qwen3.8-Max	19.2	3
Qwen3.8-27B	16.9	5
GPT-5.6-Sol	11.4	0
Qwen3-VL-8B	14.3	27
Qwen3-VL-32B	14.8	18

Skill and tool choices differ across models. In Figure 4, blue cells show the percentage of 112 tasks with a successful skill load; orange cells show each family’s percentage of non-finish tool calls. GPT-5.6-Sol has the largest summed skill-load frequency, including driving_control in 105 tasks, 96 at the initial destination wake, matching the guide’s navigation trigger. It also makes many navigation calls, mainly minimap and target-speed requests, yet has low arrival and completion-weighted outcomes. Qwen3.8-Max navigates effectively without loading that guide, so loading it is not required for effective tool use. Appendix B.1 gives the counts.

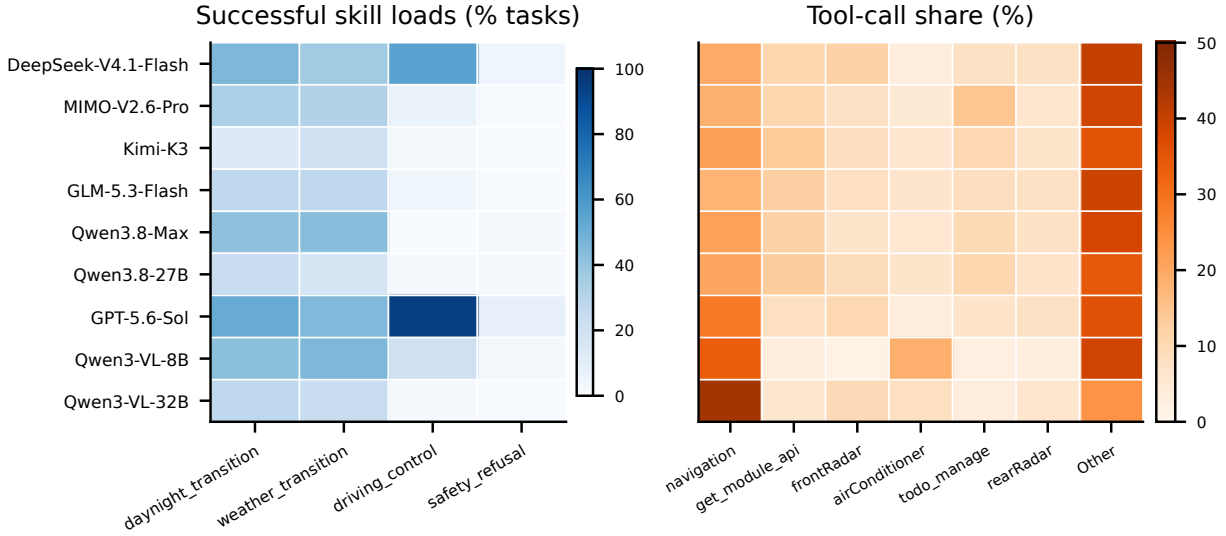


Figure 4 Ego DA skill and tool choices pooled over Basic and Multi-Agent tasks. Both heatmaps share the model order shown on the left. Left: fraction of tasks with a successful load of each skill. Right: share of non-`finish` tool calls in the six most common tool families; “Other” contains the remainder. Each panel has its own percentage color scale.

Table 3 Ego PA ablation using the main-table aggregation. Each scored cell reports the PA-off value and its change from PA on in parentheses; arrival changes are percentage points.

Model	Basic (80 tasks)			Multi-Agent (32 tasks)		
	Arr. (%)	Drive	Cabin	Arr. (%)	Drive	Cabin
Kimi-K3	61.3 (+10.0)	74.6 (−4.3)	71.6 (+30.1)	56.3 (+15.6)	77.2 (+5.8)	81.6 (+32.4)
GLM-5.3-Flash	62.5 (+17.5)	79.1 (+0.5)	75.6 (+20.0)	65.6 (+3.1)	78.8 (−2.1)	83.3 (+20.2)

5.4 Impact on Surrounding Vehicles

The ego vehicle’s behavior also shapes surrounding traffic. Figure 5 compares required non-ego trip vehicles, including fixed peers, with same-task SUMO runs. It shows added delay and changes in collision and non-arrival rates; Appendix A.6 defines the metrics.

Surrounding traffic bears different costs across models and task groups. Across the seven models shown, the equal-weight mean of the four axes across both groups is largest for GPT-5.6-Sol and smallest for DeepSeek-V4.1-Flash. On Basic, GPT has the largest burden, chiefly from delays and missed trips, consistent with its slow, often unfinished ego trips in Table 2; Qwen3.8-27B has the smallest. On Multi-Agent, Qwen3.8-27B has the largest burden, including the highest collision-rate increase and substantial queue delay. Qwen3.8-Max has the smallest burden and the fewest added non-arrivals despite visible delays. Although its surrounding-vehicle collision rate is lower than SUMO’s, we plot this favorable change as zero collision excess because the radar emphasizes the adverse effects that predominate across models.

5.5 Ablation

We disable the ego vehicle’s Personal Agent and Passenger Judge while retaining both components for the fixed peer agents. Table 3 reports the resulting changes. Disabling ego PA raises overall arrival by 11.6 percentage points for Kimi-K3 (48.2% to 59.8%) and 13.4 points for GLM-5.3-Flash (50.0% to 63.4%). Across both task groups, field-weighted Cabin scores increase by 30.9 and 20.2 points, respectively.

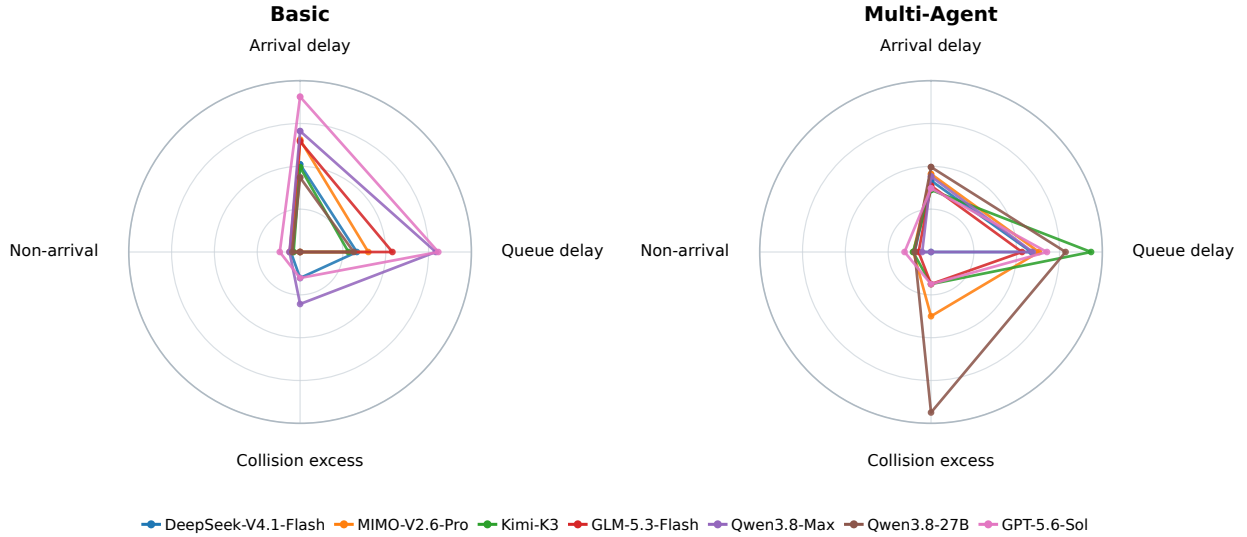


Figure 5 Surrounding-traffic burden relative to matched SUMO runs in Basic (left) and Multi-Agent (right). Axes show added delay and collision/non-arrival rate excess; reductions appear as zero. Appendix A.6 gives definitions and scales.

In-cabin requests compete with time-critical tasks. With PA enabled, ego acts on passenger device requests while missing simultaneous mandatory cabin checks in 23/112 Kimi-K3 tasks (20.5%) and 21/112 GLM tasks (18.8%). This exposes difficulty coordinating concurrent obligations.

Passenger preferences versus trip completion. Among tasks completed only with ego PA disabled, 9/21 for Kimi-K3 (42.9%) and 10/26 for GLM (38.5%) show a low-speed passenger request, matching speed command, and PA-on timeout without collision or route failure. Table 12 in Appendix C gives examples, suggesting passenger preferences can delay arrival.

6 Conclusion

VehicleArena studies independent agents whose private plans meet in a shared physical world with persistent, observable consequences. Its 3D traffic world, passenger requests, cabin actions, and non-compensatory metrics connect local decisions to system-level effects. Across nine models, strong passenger-request or cabin scores often coexist with failed trips, while the evaluated driving policies can reduce other vehicles’ arrival rates. Future work should test longer episodes, richer sensing, broader road cultures, and transfer to real driving data.

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A Experimental Settings

A.1 Inference Configuration

Role-specific inference. Table 4 summarizes the requested inference settings and documented thinking/effort controls. Effort labels are model-specific; N/A marks controls not exposed by the model. Model specifications are listed in Appendix G.

Table 4 Recorded inference settings for the main comparison. Context budgets are in tokens; 32k denotes 32,768 output tokens per call.

Role	Model	Temperature	Thinking	Effort	Context budget	Output limit
Ego DA	DeepSeek-V4.1-Flash	0.7	Enabled	max	1000000	32k
	MiMo-v2.6-Pro	1	Enabled	N/A ^a	1000000	32k
	Kimi-K3	0.7	Enabled	max	1000000	32k
	GLM-5.3-Flash	0.7	Enabled	max	1000000	32k
	Qwen3.8-Max	0.7	Enabled	xhigh	1000000	32k
	Qwen3.8-27B	0.7	Enabled	xhigh	1000000	32k
	GPT-5.6-sol	0.7	Enabled	high	1000000	32k
	Qwen3-VL-8B	0.7	N/A ^b	N/A	1000000	32k
	Qwen3-VL-32B	0.7	N/A ^b	N/A	1000000	32k
Peer DA	Qwen3.8-27B	0.7	Enabled	xhigh	1000000	32k
PA	Qwen3.8-27B	0.7	Enabled	xhigh	1000000	32k
Judge	Qwen3.8-27B	0.0	Enabled	xhigh	1000000	32k

^aMiMo does not expose adjustable effort. ^bQwen3-VL uses instruct checkpoints.

In MULTILLM, peer vehicles retain their distinct personality prompts and active PAs and Judges in both the oracle baseline and the evaluated run. Only the ego vehicle changes controller; all other experimental settings are matched. The ego PA/Judge ablation disables those two ego roles while retaining the fixed peers’ configuration. Without the ego PA, no ego passenger requests are generated. Cabin scoring uses the same automatic rule path in both settings (Section E.2).

A.2 Task Split

Split and counting conventions. Table 5 summarizes 100 BASIC training tasks and 112 test tasks: 80 BASIC and 32 MULTILLM tasks selected by the frozen run manifests. The Basic training pool contains the 100 scenarios outside the Basic test manifest. The table counts all LLM-controlled vehicles, including peers in each MULTILLM task. Map counts are deduplicated within each split, so aggregate map counts are not sums of the component splits.

Task families. BASIC covers crosswalk yielding, signalized and unsignalized intersections, straight following, required and continuous turns, full-network navigation, lead-vehicle braking, narrow-road interaction, red-light stopping, oncoming and crossing streams, unprotected left turns, obstacle-driven gap changes, platoon pressure, and merge streams. MULTILLM combines narrow meetings, unprotected left turns, synchronized four-way intersections, multi-vehicle merges, and dense four-way and merge variations. Across both splits, weather can be stable or transition among cloudy, foggy, rainy, snowy, heavy-rain, heavy-snow, and hail

Table 5 Task-suite statistics. Tasks and maps are counts; other entries are per-task means, with ranges in parentheses.

Statistic	Train	Test BASIC	Test MULTILLM	Overall
Tasks	100	80	32	212
Maps	57	48	13	89
LLM vehicles	1.0	1.0	4.0	1.5
NPC vehicles	13.1 (8–24)	13.2 (8–24)	10.3 (8–13)	12.7 (8–24)
Pedestrians	0.5 (0–8)	0.5 (0–8)	0.0	0.4 (0–8)
Ego route (m)	373.8 (22.0–2620.7)	402.4 (63.4–4867.2)	199.6 (75.8–427.9)	358.3 (22.0–4867.2)
Time limit (s)	88.0 (20.3–514.3)	89.8 (21.7–490.6)	50.0 (24.6–73.9)	82.9 (20.3–514.3)

Ego route is the lane-level distance from the initial pose to the destination, using the frozen route or the lane planner when no initial route is stored. Time limit is the calibrated simulation deadline in Equation 1.

conditions; illumination includes dawn, morning, noon, afternoon, dusk, and night. Passenger requests are generated online by PA under the configuration described in Section A.3.

A.3 PA and Judge Scheduling

A.3.1 Personal-Agent wakes

The PA input includes passenger-visible state, recent motion, the wake category, prior requests, and passenger-visible DA replies. Judge grades, hidden evaluation state, and DA private reasoning are excluded.

Table 6 Default PA scheduling. The random stream is independently seeded per vehicle and scenario.

Wake source	Default rule
Shared passenger-visible events	Simulation start, acoustic cue, weather change, and day–night change.
Hard braking edge	Acceleration at most -3 m/s^2 for 0.3 seconds; detection is rearmed only after acceleration exceeds -1.5 m/s^2 .
Prolonged-stop edge	Speed below 0.5 km/h for 15 seconds; detection is rearmed after speed exceeds 2 km/h.
Independent random wake	A delay sampled from $U(15, 45)$ simulation seconds, unaffected by the Driving-Agent heartbeat.
Coalescing	A five-second cooldown merges repeated events by category. A PA may also finish without issuing a request; its update still wakes the Driving Agent once at that boundary.

Judge checks, tool receipts, request expiry or non-completion, and DA heartbeats do not generate PA wakes.

A.3.2 Typed deferred-request triggers

Table 7 lists the supported trigger families. The runtime filters candidates using the remaining route, scheduled environmental events, and free-flow time available for the Judge window. Numeric parameters are bounded by the current state; free-text predicates are not supported.

Table 7 Declarative Judge-trigger catalog. Only currently feasible, parameter-bounded members of this catalog are shown to the PA.

Family	Available predicates
Time	A bounded delay after request creation.
Route and map	Approaching, entering, or exiting the next intersection; distance to destination below a bounded threshold.
Vehicle state	Vehicle stopped, vehicle resumed moving, or a speed threshold held for a bounded duration.
Environment	A scheduled weather transition to an exposed condition, or a scheduled transition to darkness.

Selected predicates are checked at the 0.1-second physics resolution.

A.3.3 Request lifecycle and Judge windows

The request contract records core and secondary criteria, a one-shot or ongoing kind, expected response time, validity duration, and an optional trigger. Judge checks occur at offsets of 0.1, 1, and 3 seconds after immediate creation or trigger activation, with at most three checks per phase. Intermediate checks retain criterion status; the final verdict assigns one grade. Evidence includes the contract, DA replies, tool receipts, before/after state, and the physical trace.

Table 8 Request lifecycle. Terminal states are reported as coverage rather than silently folded into the request score.

State	Meaning
Created / waiting	The frozen request exists; a deferred request is waiting for a feasible selected predicate.
Activated / checking	The predicate occurred and the Judge gathers the bounded time-window evidence.
Completed	Evidence establishes fulfillment within the request validity period; one-shot requests may close early once all core criteria are established.
Uncompleted / unverified	The check budget ends with unmet criteria or insufficient evidence, respectively.
Superseded / episode ended	A later PA request replaces the request, or the physical episode ends before its pending lifecycle is resolved.
NA / infrastructure failure	An invalid request or insufficient evidence is unscored (NA); operational failures are tracked separately.

Ongoing requests remain pending until the final observation window. Verdict validation checks criterion status, phase boundaries, timestamps, and trace consistency; malformed verdicts remain evaluation failures.

Algorithm 1 Passenger Request and Verification Loop at Each World Boundary

Require: World state s , events e , active request sheet S , recorded execution history

```
1:  $p \leftarrow \emptyset$  ▷ only passenger updates are returned to DA
2: if PAIsDue( $s, e$ ) then
3:    $\Omega \leftarrow \text{AvailableTriggers}(s, \text{RemainingRoute}, \text{TimeBudget})$ 
4:    $d \leftarrow \text{PA}(\text{PassengerView}(s), S, \Omega)$ 
5:   Validate structured request updates and freeze their acceptance criteria
6:    $(S, p) \leftarrow \text{ApplyAcceptedUpdates}(S, d)$ 
7: end if
8: for each pending request  $r \in S$  do
9:   Update  $r$ 's verification schedule from its trigger, deadline, and episode status
10:  if CheckDue( $r$ ) then
11:     $x \leftarrow \text{CollectExecutionEvidence}(r)$ 
12:     $v \leftarrow \text{VerifyRequest}(r, x, \text{PriorChecks}(r))$ 
13:    Record a valid verdict, or retain an explicit unscored status
14:    Close  $r$  if complete or at its final check; otherwise retain it for a later check
15:  end if
16: end for
17: return  $p$  ▷ no driver wake is generated by verification
```

A.4 Evaluation Metric Definitions

Delay increases are summed per vehicle, so time saved by one vehicle does not offset delays imposed on another. Trip-completion delay is computed only for vehicles that arrive in the evaluated run. Untriggered or unverified requests and failed Judge calls are reported separately, not assigned grades.

Table 9 VehicleArena metrics. Metrics are reported separately rather than combined into one scalar.

Metric	Unit and denominator	Definition and companion report
Trip completion A	Percentage of tasks	Ego arrival by T_{limit} without collision or route failure. Peer arrivals are reported separately, not required for ego success.
Driving quality D	Score in $[0, 100]$ per evaluated vehicle	Starts at 100 with deductions for driving violations; an at-fault collision or red-light violation sets the score to zero.
Passenger fulfillment R	Score in $[0, 100]$ over valid scored requests	A–F Judge grades mapped to 100, 80, 60, 40, 20, and 0. Request-level and scenario-level means use explicit denominators.
Cabin compliance C	Score in $[0, 100]$ over checked fields	Field-weighted accuracy pooled across ego checkpoints, including sentinel checks. Rules requiring unavailable equipment are excluded.
Externality E	Counts and vehicle-seconds per valid pair	Changes in NPC collisions and non-arrivals, plus positive increases in queueing time and trip-completion time relative to SUMO.
Efficiency K	Input and output tokens per task	Total model-token use, reported separately for input and output.

A.5 Passenger Evaluation and Reporting

A.5.1 Aggregation and unscored cases

For each aggregate, report its valid observation count and the numbers of unavailable, unscored, or invalid cases. An empty applicable set is reported as unavailable, not as a zero score.

The request-level mean pools all valid scored requests. The scenario-level mean first averages valid request scores within each scenario, then averages over scenarios with at least one scored request. Thus, the latter gives each applicable scenario equal weight rather than weighting it by request count.

A.5.2 Passenger-request grades

Table 10 expands the six grades in Table 9. Judge assesses only the frozen passenger requirements, not unrelated driving quality or traffic impact. Timing is based on evidence of fulfillment, not the time at which Judge returns its verdict.

Table 10 Passenger-request grading rubric. NA is an unscored status, not a seventh grade.

Grade	Score	Evidence-based interpretation
A	100	All requirements are resolved correctly within the expected response time.
B	80	All requirements are resolved, but after the expected response time and before expiry.
C	60	All core requirements are resolved, with minor secondary omissions or missing secondary evidence.
D	40	Useful partial fulfillment of core requirements, or failure to maintain a sustained requirement.
E	20	Relevant physical action is observed, but it produces no useful requested outcome.
F	0	No relevant action, an unrelated action, or an outcome contrary to the request.
NA	—	An invalid request or evidence insufficient for a fair judgment; excluded from score averages.

A requirement is resolved by observed fulfillment or a clear, truthful refusal supported by evidence that the vehicle lacks the requested capability. Such a refusal does not establish physical completion. Promises, Todo edits, and queued commands alone do not establish fulfillment; an already satisfied state can count. Sustained requirements are checked across the observation window.

A.5.3 Traffic Interaction Set

The traffic-interaction set contains all required non-ego trip vehicles, including fixed peers in MultiLLM. Evaluated and SUMO reference runs are paired by task and vehicle ID. Queueing delay, collision, and non-arrival measures use this full set; trip-completion delay is computed only for vehicles arriving in both runs. Figure 5 uses the same scope, with the measures defined in Section A.6.

A.6 Surrounding-Vehicle Impact Measures

Figure 5 uses all required non-ego trip vehicles, including fixed peers in Multi-Agent tasks. For each model and task group, let $\mathcal{T}$ contain the 80 Basic or 32 Multi-Agent tasks, and let V_t be the surrounding trip vehicles in task t . We pair each evaluated run M with its SUMO reference S by task and vehicle ID. Basic uses an all-SUMO reference; Multi-Agent uses a SUMO ego vehicle with fixed peers.

Let a_{tv}^X be arrival time, q_{tv}^X queue-wait time, and $r_{tv}^X \in \{0, 1\}$ the arrival indicator for vehicle v in run $X \in \{M, S\}$. Let $c_{tv}^X \in \{0, 1\}$ indicate whether that vehicle was involved in at least one collision in run X . Define $V_t^\cap = \{v \in V_t : r_{tv}^M = r_{tv}^S = 1\}$, $N_{\text{trip}} = \sum_{t \in \mathcal{T}} |V_t|$, and $[x]_+ = \max(x, 0)$. The four baseline-relative

measures are

$$\begin{aligned}
D_{\text{arr}} &= \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{v \in V_t^{\cap}} [a_{tv}^M - a_{tv}^S]_+, & D_{\text{queue}} &= \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{v \in V_t} [q_{tv}^M - q_{tv}^S]_+, \\
\Delta C &= \frac{100}{N_{\text{trip}}} \sum_{t \in \mathcal{T}} \sum_{v \in V_t} (c_{tv}^M - c_{tv}^S), & \Delta N &= \frac{100}{N_{\text{trip}}} \sum_{t \in \mathcal{T}} \sum_{v \in V_t} (r_{tv}^S - r_{tv}^M).
\end{aligned} \tag{2}$$

The delays D_{arr} and D_{queue} have units of vehicle-seconds per task. ΔC and ΔN are treatment-minus-SUMO changes in collision and non-arrival rates, in percentage points. A vehicle counts once in ΔC even if it has multiple collision events. The radar axes plot $(100D_{\text{arr}}/45, 100D_{\text{queue}}/60, 100[\Delta C]_+/1.25, \Delta N)$. The delay divisors are 45 and 60 vehicle-seconds per task; the collision divisor is 1.25 percentage points. Because the radar cannot show a negative radius, a collision-rate decrease is displayed at zero on that axis; the signed difference remains in the underlying data. Non-arrival is shown as the baseline difference in percentage points without further scaling. For descriptive comparisons, we average these four plotted coordinates and then average the Basic and Multi-Agent values with equal weight; this is a visual summary, not an additional evaluation metric.

A.7 Completion-Weighted Outcome Index

For each model and task group, A , D , R , and C denote the Arr. (%), Drive, Req., and Cabin metrics in Table 1, respectively. We use their unrounded values and write A as a percentage (e.g., $A = 65\%$). The plotted index is

$$I = A \frac{D + R + C}{3}. \tag{3}$$

B Supplementary Analyses

B.1 Model-Level Skill, Tool, and Token Counts

Table 11 gives the counts behind Figure 4 and the ego DA token totals for the same models. Each row pools 80 Basic and 32 Multi-Agent tasks. Skill entries count tasks with a successful load of the named skill; tool entries count non-finish calls to each tool family. Day/night, Weather, Driving, and Safety denote `daynight_transition`, `weather_transition`, `driving_control`, and `safety_refusal`. The tool columns Nav., API, Front, A/C, Todo, and Rear denote `navigation`, `get_module_api`, `frontRadar`, `airConditioner`, `todo_manage`, and `rearRadar`, respectively. Other collects the remaining tool families. Token entries are total ego DA input and output tokens across the 112 tasks, reported in millions (M; 1 M = 10^6 tokens) and rounded to two decimal places.

B.2 Token Cost Analysis

For the cost comparison in Figure 6, we count ego DA input and output tokens per task. Passenger agents, judges, and fixed peer models are excluded. Both panels use the same model order, set by Basic token use, and the same scales.

Token totals add a different view of effort. Runs with similar wake and tool counts can process different amounts of text during each wake, so interaction counts need not predict inference use. We therefore compare tokens with the same completion-weighted outcome to see whether additional processing is accompanied by better task results.

On Multi-Agent, GLM-5.3-Flash substantially outperforms Kimi-K3 and Qwen3.8-27B at nearly the same token budget, while Qwen3.8-Max reaches the highest outcome with only moderately more tokens. At a higher, roughly two-million-token budget, DeepSeek-V4.1-Flash outperforms GPT-5.6-Sol. Thus, token volume alone does not explain task success. We use token count as a proxy for inference cost without accounting for differences in provider pricing.

Table 11 Per-model skill loads, tool calls, and ego DA token counts.

Successful skill loads (tasks)							
Model	Day / night		Weather		Driving		Safety
DeepSeek-V4.1-Flash	51		41		62		5
MIMO-V2.6-Pro	37		35		7		0
Kimi-K3	14		23		2		0
GLM-5.3-Flash	30		30		4		0
Qwen3.8-Max	46		48		0		1
Qwen3.8-27B	26		19		1		1
GPT-5.6-Sol	57		50		105		9
Qwen3-VL-8B	47		51		23		2
Qwen3-VL-32B	30		26		1		0
Tool-family calls							
Model	Nav.	API	Front	A / C	Todo	Rear	Other
DeepSeek-V4.1-Flash	2,717	1,467	1,669	442	1,085	1,093	5,681
MIMO-V2.6-Pro	1,658	953	687	467	1,329	563	3,549
Kimi-K3	1,681	1,046	645	439	787	513	2,747
GLM-5.3-Flash	1,289	897	588	458	624	562	2,848
Qwen3.8-Max	1,772	1,000	559	461	817	624	3,225
Qwen3.8-27B	1,336	859	588	426	685	450	2,248
GPT-5.6-Sol	4,495	1,286	1,627	544	1,052	1,176	5,653
Qwen3-VL-8B	5,243	525	209	2,898	356	443	6,065
Qwen3-VL-32B	7,877	993	1,674	1,425	575	1,067	4,235
Ego DA tokens (M)							
Model	Input			Output			Total
DeepSeek-V4.1-Flash	331.64			43.89			375.53
MIMO-V2.6-Pro	162.80			15.63			178.43
Kimi-K3	117.53			7.69			125.22
GLM-5.3-Flash	121.58			8.73			130.31
Qwen3.8-Max	141.16			12.25			153.41
Qwen3.8-27B	108.83			8.86			117.68
GPT-5.6-Sol	265.82			8.94			274.76
Qwen3-VL-8B	234.35			5.89			240.24
Qwen3-VL-32B	301.43			6.17			307.60

Figure 6 shows that similar token budgets can yield very different outcomes. On Basic, Qwen3.8-Max achieves the strongest outcome among models using roughly one to two million ego DA tokens per task; it also matches DeepSeek-V4.1-Flash while using less than half as many tokens.

C Passenger Requests in the PA Ablation

Table 12 expands four representative ego trajectories into six individual passenger requests, paired with the road and environment context at the time of each request. Times are simulation seconds. Requests are translated or abridged, with unrelated device clauses omitted. The numerical speeds are passenger

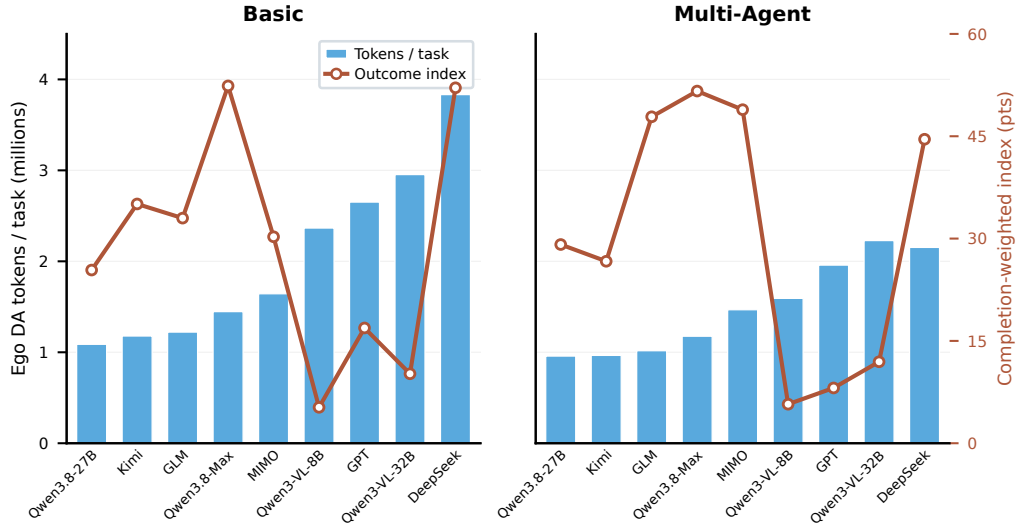


Figure 6 Inference cost and task outcome by model (Basic: left; Multi-Agent: right). Bars show mean ego DA tokens per task (left axis, millions); the line shows the completion-weighted index (right axis, points).

preferences, not road speed limits; distance thresholds specify when the passenger wants an action to occur, not the measured remaining distance when the request is issued.

Table 12 Scene context and low-speed passenger requests in the PA ablation.

Scene at request time	Passenger request
Urban lead-vehicle following in Beijing, in cloudy weather. At 5.0 s, a hard-braking event prompts a request for more separation from the car ahead.	After that hard brake, slow to about 25 km/h so we keep more distance from the car ahead.
The same following episode, after rain begins. At 20.4 s, the passenger updates rain-related device settings and plans a quiet final approach.	Once we are within about 130 m of the destination, slow to about 20 km/h for the final approach and turn the music off.
Platoon-following traffic in Shanghai. At 14.1 s, fog appears as afternoon changes to dusk, reducing visibility.	It is foggy and we are approaching the destination; slow to about 20 km/h.
The same foggy episode at dusk. At 19.1 s, a later request specifies a further slowdown for the final approach and preparation for arrival.	When navigation shows less than 70 m to the destination, slow to about 10 km/h and pause the music to prepare for arrival.
An urban crosswalk route in Shanghai at dusk. Rain begins at 15.1 s; the passenger requests wipers, fog lights, and a cautious final approach.	Once we are within 50 m of the destination, slow to below 15 km/h, then stop smoothly after arrival.
An urban road with oncoming traffic in Sydney, in the afternoon. At 10.0 s, a passenger who had been reading asks for quieter audio and prepares to pack up.	Within 100 m of the destination, approach gently at under 10 km/h and turn off the passenger reading light so I can pack up.

D Driving-Rule Scoring Details

Driving quality starts at 100. An at-fault collision or red-light violation sets the score to zero; otherwise, the score is $\max(0, 100 - \sum_j p_j)$, where p_j is a recorded deduction in Table 13. These are event-level penalties, not weights of a normalized average. Weather-dependent cabin compliance is scored separately rather than deducted again here.

Table 13 Driving-score deduction schedule. Repeated detections are grouped into events or episodes before scoring.

Violation	Deduction
At-fault collision or red-light violation	Score becomes 0
Sustained speeding, at least 5 seconds	20
Short speeding episode, 1 to less than 5 seconds	5
Unjustified stopping	5 each
Unjustified delay in starting after a green light	5
Lane change without signaling	10
Turn without signaling	10
Failure to cancel a turn signal	2
Unsafe lane change	10
Late response to a lead-vehicle, junction, or pedestrian risk	10 each
Vehicle near miss	15
Junction-conflict near miss	15
Pedestrian near miss	20
Blocking an intersection	10
Harsh acceleration	3
Unnecessary harsh braking	3
High longitudinal jerk	1
Unnecessary horn use	2
High-beam misuse	2
Entering a closed road	10

Deduction types have a three-second cooldown. Continuous unjustified stops and intersection blocking incur an initial penalty after three seconds and repeat at five-second intervals while the condition persists.

Driving deductions use observed physical behavior. Pedestrian near misses are identified from vehicle-pedestrian body clearance and vehicle speed, rather than predicted arrival timing alone. Late responses and near misses within one continuous pedestrian encounter incur only the most severe deduction, preventing repeated penalties for the same encounter.

E Vehicle Modules and Configuration

The task configuration resolves a separate immutable capability set for every vehicle before the first simulation step. Table 14 lists the supported fields. A scenario may provide defaults for all vehicles and overrides for an individual vehicle. Validation occurs before the episode starts, so an unknown module or invalid chassis value cannot become a mid-episode tool failure.

The same resolved capability set controls module instantiation, the initial tool catalog, the available skill catalog, and physical limits. If DA tries to inspect or invoke an uninstalled module, the interface returns an explicit `capability_not_available` result. This makes an unsupported request distinguishable from a failed attempt to use an installed device.

Table 14 Per-vehicle configuration fields.

Field	Effect
Equipment profile	Selects a named set of cabin and sensor modules.
enable_modules / disable_modules	Adds or removes named optional modules for one vehicle. Required world and driving modules cannot be removed.
Chassis profile	Selects vehicle length, width, acceleration, braking, and lane-change duration.
Chassis overrides	Replaces individual positive chassis values for a task-specific vehicle variant.
Trusted extension	At run initialization, registers additional modules, equipment profiles, chassis profiles, scenario layers, or rule directories. Scenario data cannot import code.

E.1 Built-in module catalog

Table 15 lists the 40 modules registered by the current release. These are the modules that a profile may expose; an individual vehicle receives only the subset selected by its capability set. The five world and driving modules are required by the runtime. The others are optional and can be included, excluded, or supplemented by a trusted extension.

Table 15 Built-in VehicleArena modules. Names are the module identifiers used by the capability resolver and tool-discovery interface.

Group	Registered modules
World and driving	navigation, map, weather, dayNight, speedLimit
Sensors	frontRadar, rearRadar, lidar
Climate and comfort	airConditioner, seat, sunshade, readingLight
Display, media, and communication	bluetooth, broadcast, centerInformationDisplay, conversation, HUD, instrument- Panel, music, overheadScreen, radio, video, horn
Lighting and safety	fogLight, hazardLight, highBeamHeadlight, lowBeamHeadlight, positionLight, tailLight, turnSignal, wiper
Body and physical controls	door, footPedal, frontTrunk, fuelPort, rearviewMirror, steeringWheel, sunroof, trunk, window

E.2 Declarative rule format and evaluation

Cabin requirements are authored as validated YAML rules rather than embedded in evaluator code. Each rule has an identifier and one domain-specific trigger, an optional guard, expected actions, and optional tolerances. The supported domains are weather transitions, day–night transitions, map events, and user intent. Tolerances can name accepted alternatives, fields to ignore, a required numeric trend, or a numerical ceiling. Negative checks specify a state that must not occur. Unknown fields or invalid action syntax are rejected when the rule files are loaded.

In the reported experiments, automatic Cabin expectations are derived from weather, day–night, and map transitions. Although the schema supports user intent, PA-generated requests are evaluated separately by the Passenger Judge and do not generate user-intent Cabin checkpoints in either PA-on or PA-off. Disabling the ego PA/Judge removes ego request generation and judgment; fixed peers retain both roles.

At a relevant transition, the rule engine derives the expected action set from the current and previous world snapshots. For scoring, those actions run on an isolated reference VehicleWorld. The cabin evaluator compares the reference state change with the actual state change of the evaluated vehicle, checking the target fields affected by the reference actions with the declared tolerances. Rules requiring uninstalled modules are excluded. An already satisfied expectation counts as one fulfilled check. A newly violated negative invariant,

or a checkpoint for which all reference candidates fail to execute, contributes one failed check. The reported Cabin score is $100 \frac{\sum \text{correct_fields}}{\sum \text{total_fields}}$, pooled across the ego checkpoints in the reported task set. The number of checked fields can vary between PA-on and PA-off because reference-action state changes depend on the current device state. Free-form DA text and an API call without its required state change do not count as compliance.

F Supported Cities

Table 16 lists map areas represented in the configured scenario catalog by region and city. Evaluation tasks use a subset of these areas.

Table 16 City-level map inventory. Each local area corresponds to one lane-level map; the table does not imply that every map appears in the 112-task evaluation set.

Region	City	Local area(s)
Mainland China (57)	Beijing	Guomao; Sanlitun; Tiananmen; Wangjing; Wudaokou; Xidan; Yizhuang; Zhongguancun
	Changchun	Chaoyang
	Changsha	Furong; Wuyi
	Changzhou	Tianning
	Chengdu	Chunxi; Gaoxin; Jinjiang
	Chongqing	Jiefangbei
	Dalian	Zhongshan
	Dongguan	Nancheng
	Foshan	Chancheng
	Fuzhou	Wuyi
	Guangzhou	Panyu; Tianhe
	Guiyang	Guanshanhu
	Haikou	Longhua
	Hangzhou	Binjiang; Xihu
	Harbin	Zhongyang
	Hefei	Shushan; Zhengwu
	Hohhot	Saihan
	Jiaxing	Nanhu
	Jinan	Quancheng
	Kunming	Cuihu
	Lanzhou	Chengguan
	Lhasa	Chengguan
	Nanchang	Honggutan
	Nanjing	Hexi; Xinjiekou
	Nantong	Chongchuan
	Ningbo	Tianyi
	Qingdao	Shinan; Wusi
	Shanghai	Hongkou; Jing'anbei; Lujiazui; Pudong Zhangjiang
	Shenyang	Heping; Zhongjie
	Shenzhen	Futian; Nanshan
	Suzhou	Guanqian
	Taiyuan	Yingze
	Tianjin	Heping
	Wuhan	Hankou

Continued on next page.

Table 16 continued.

Region	City	Local area(s)
	Wuxi	Taihu
	Xiamen	Huli
	Xi'an	Zhonglou
East and Southeast Asia (10)	Bangkok	Silom
	Hanoi	Hoankiem
	Hong Kong	Central
	Jakarta	Central
	Kuala Lumpur	Bukit
	Osaka	Namba
	Seoul	Gangnam
	Singapore	Orchard
	Taipei	Xinyi
	Tokyo	Shinjuku
Europe (13)	Amsterdam	Centrum
	Berlin	Mitte
	Helsinki	Keskusta
	Istanbul	Beyoglu
	London	West End
	Madrid	Centro
	Moscow	Tverskaya
	Paris	Champs
	Prague	Stare Mesto
	Rome	Centro
	Stockholm	Norrmalm
	Vienna	Innere
	Warsaw	Srodmiemie
North America (5)	Chicago	Loop
	Los Angeles	Downtown
	New York	Manhattan Midtown
	San Francisco	SoMa
	Toronto	Downtown
Oceania (2)	Melbourne	CBD
	Sydney	CBD
Middle East and Africa (2)	Cairo	Downtown
	Dubai	Downtown

G Model Specifications

Table 17 summarizes the nine models evaluated in our experiments. Peer DA, PA, and Judge all use Qwen3.8-27B.

Table 17 Specifications of the large language models evaluated in our experiments.

Models	#Para	Launch date	Context (tokens)	Scaling	Corporation	License
DeepSeek-V4.1-Flash	748B	2026-09-10	1,048,576	Effort	DeepSeek	MIT
MiMo-v2.6-Pro	1.02T	2026-09-22	1,048,576	Budget	Xiaomi	MIT
Kimi-K3	2.8T	2026-07-17	1,048,576	Effort	Moonshot AI	Kimi K3
GLM-5.3-Flash	320B	2026-09	1,048,576	Effort	Z.ai	MIT
Qwen3.8-Max	2.4T	2026-08-02	1,000,000	Effort	Alibaba	Proprietary
Qwen3.8-27B	27B	2026-08-14	262,144 ^a	Effort	Alibaba	Apache 2.0
GPT-5.6-sol	—	2026-07-09	1,050,000	Effort	OpenAI	Proprietary
Qwen3-VL-8B	8.8B	2025-10-15	262,144 ^a	Budget	Alibaba	Apache 2.0
Qwen3-VL-32B	33.4B	2025-10-21	262,144 ^a	Budget	Alibaba	Apache 2.0

^aNative context, extensible to 1M tokens. Context is model capacity; the experimental output limit is 32,768 tokens (Table 4). Effort/Budget denotes reasoning-effort/generation-length control. Qwen3-VL uses Instruct checkpoints; counts include the vision encoder. Licenses apply to released weights where available; Qwen3.8-Max is the hosted model.

H Prompts

This section presents the prompt templates used by the Driving Agent, Personal Agent, and Judge in VehicleArena.

Driving Agent Prompt

SYSTEM

You directly control one vehicle in a continuous road-world simulation. Use tools for driving and cabin operations. The world advances in 0.1-second steps. A target-speed command persists; accepted motion commands are not completed maneuvers. Use CameraVisual for visible traffic and signal state. Use installed radar and LidarBEV only when available. The assigned destination remains your long-term Todo goal. Route planning supplies guidance, not automatic route following. Choose legal junction maneuvers and signal lane changes yourself. Use `get_module_api` and `load_tools` for installed capabilities. Use `set_heartbeat_interval` or `schedule_next_wake` to observe again. Call `finish` when this wake is complete.

<installed-module catalog and chassis limits>

<available skill catalog>

<multi-vehicle context: vehicle_id>

<driving-role addendum: driver_prompt, if configured>

USER AT EACH WAKE

CurrentWake: <time, ego state, active control, wake policy,

Todo board, passenger updates, new events>

CameraVisual: <aligned cockpit image>

LidarBEV: <image only if LiDAR is installed>

LoadedCapabilities: <startup or changed callable schemas>

PreviousWake: <at most one retained exchange if budget permits>

Personal Agent Prompt

SYSTEM

You are the human passenger riding in this vehicle. React only to passenger-observable cabin, weather, motion, traffic, comfort, and safety state. Do not control equipment directly or request a change to the assigned destination. Call finish when no request is warranted. Otherwise submit one natural passenger utterance with explicit, observable core outcomes and genuinely optional secondary outcomes. Use `send_immediate_request` for outcomes beginning now. Use `send_triggered_request` only for a two-stage request with immediate and triggered outcomes and an offered judge trigger. Give `request_kind`, `expected_response_s`, and `valid_for_s`. If an active sheet is replaced, restate every still-desired outcome; cancel it only when none remains wanted.

USER AT PA WAKE

```
{  
  "type": "personal_agent_wake",  
  "persona": <passenger persona>,  
  "todo": {"long_term_goal": <passenger goal>},  
  "observation": <passenger-visible state, active request sheet,  
    request design, available judge triggers>  
}
```

TOOLS

`send_immediate_request` | `send_triggered_request` |
`cancel_passenger_request` | `finish`

Judge Prompt

SYSTEM

You are an independent VehicleArena passenger-request judge.

<passenger grade rubric and physical-outcome contract>

Judge only the frozen requested outcomes using observed evidence. An accepted command, Todo edit, or promise alone is not proof of physical fulfillment. Previously committed control may explain observed motion. Classify each core and secondary criterion as met, unsupported_refused, unmet, or unverified. Use `unsupported_refused` only for an explicit passenger-facing refusal backed by authoritative unavailability evidence. For triggered criteria, measure time from trigger activation; for immediate criteria, measure time from request creation. At an intermediate window, submit criterion statuses without an A--F grade. At the final window, submit one evidence-backed grade or explicit NA status and fulfillment times for resolved phases. Judge ongoing behavior within the observed window.

USER AT JUDGE CHECK

<frozen request and criteria; phase/timing metadata; DA replies;
tool receipts; before/after device state; physical trace;
prior checks; intermediate or final window flag>

TOOL AT INTERMEDIATE WINDOW: `submit_passenger_check`

TOOL AT FINAL WINDOW: `submit_passenger_judgement`

I Skills

Skills are optional Markdown guides discovered by DA through the catalog embedded in its system instruction. Calling `load_skill(skill_name)` loads the named guide into retained context. The shipped catalog contains the four guides in Table 18. Guidance naming equipment absent from the current vehicle is filtered when loaded. The weather guide expands a shared set of weather-safety rules at load time.

Table 18 Operational skill catalog in the implementation.

Skill	Use
driving_control	Motion commands, perception, routing, and command receipts.
weather_transition	Equipment changes for weather and visibility.
daynight_transition	Lights, displays, and mirrors around darkness.
safety_refusal	Unsafe-request refusal and safe alternatives.

The following listings show the complete guide bodies before vehicle-specific equipment filtering, with the shared weather rules expanded. YAML frontmatter is omitted; typography is normalized, and the intersection-name example is translated into English.

I.1 driving_control

Skill: driving_control

Driving Control Tool Reference

This skill explains what the driving tools do. It does not choose a driving personality, desired speed, risk tolerance, following distance, signal compliance, or maneuver for the driver.

Continuous execution

- The physical world advances in 0.1-second steps.
- A successful control call means the command was accepted for the next world commit. It does not mean the maneuver completed instantly.
- Target speed persists until replaced.
- Commands submitted in one wake are committed as one batch. Commands that control the same motion slot do not stack: the later command wins and the earlier command receipt reports `superseded`.
- Acceleration and braking are bounded.
- A lane change is a continuous lateral trajectory lasting several seconds.
- Intersections use explicit source-lane -> connector -> destination-lane paths.
- Vehicle rectangles and pedestrian circles are checked with swept collision geometry throughout each step.

Perception

Every vehicle wake includes `CameraVisual`, captured from the synchronized Web3D cockpit. It is the optical source for visible lanes, road users, signal lights and vehicle lamps, and is affected by weather, daylight and lighting. An installed optional LiDAR adds `LidarBEV`: the established ego-centred rule-rendered top-down geometry image. It deliberately omits traffic-light state, lamp effects, weather and day/night appearance. Vehicles without LiDAR receive no substitute image. Installed front/rear millimetre-wave radar tools provide numeric anonymous vehicle tracks. The visual

Skill: driving_control

heartbeat defaults to one second; the driver may set it between 0.1 and 30 simulation seconds. Genuine onboard radar warnings may interrupt it; evaluator-only world-risk labels never wake the driver.

Route planning

``navigation_route_plan(destination)`` resolves a destination to a suggested lane route for display. It returns only a compact success receipt; inspect the route geometry with ``navigation_minimap``. Planning neither moves the vehicle nor selects a SUMO intersection connector.

The mini-map is heading-up with the ego vehicle near the lower centre. ``scope="route"`` provides a wider forward driving window and ``scope="local"`` provides a closer junction/lane window. Long routes leave the image boundary; they are not compressed into a whole-trip overview.

Destination names may identify:

- an intersection, such as ``"Road A & Road B"``;
- a POI or landmark on a road.

Arrival is committed by the physical world after contact/collision checks, not merely because a tool was called.

For an intended left/right/U-turn, call ``navigation_select_maneuver("left" | "straight" | "right" | "u_turn")`` before entering the junction. An explicit straight selection is also supported. One successful call selects exactly one legal source-lane -> connector -> destination-lane movement. A direction unavailable from the current lane is rejected; change lanes explicitly and wait for completion before trying again. The selection remains active through that junction and is not cancelled by an intermediate wake. After exiting that junction, the selection is consumed; a left turn does not make the vehicle keep choosing left at later junctions.

Without an explicit selection, the vehicle continues through the current lane's unique legal straight connector, if one exists. This is lane continuation, not automatic following of the highlighted navigation route. If there is no unique straight connector, the system does not brake or choose a turn for you; an unresolved route endpoint can terminate the trip as a failure. Observe the road and decide in time. Background NPC routes remain fully SUMO-controlled. If the highlighted route terminates on the current lane, continue to its endpoint without selecting another junction maneuver.

Traffic signals are movement-specific: a straight green arrow does not permit a left turn whose arrow is red. Observe the signal for your intended movement.

Longitudinal commands

``navigation_set_speed(...)`` submits a persistent target speed plus optional acceleration and ordinary-deceleration limits. SUMO applies the command under the installed chassis bounds. Speed zero requests an ordinary physical stop.

``navigation_emergency_stop(reason)`` requests the installed chassis emergency braking envelope. It still takes physical time.

Skill: driving_control

There are no `follow`, `overtake` or `yield` modes. Express those decisions through explicit speed and lane-change commands, then observe their physical result.

Lane changes and U-turns

`navigation\_change\_lane("left" | "right")` starts a physical lane-change trajectory when the requested lane exists. Target-lane bodies remain present, and a collision can occur during the maneuver.

The physical actuator does not add an indicator automatically. Before starting a lane change, call the preloaded `turnSignal\_switch` tool with the matching direction. Switch it off after the maneuver finishes. Other agents can perceive the emitted indicator subject to their optical range and conditions.

`navigation\_u\_turn()` is the dedicated form of selecting a U-turn connector from the current lane. It does not change lanes, rotate, or relocate the vehicle automatically.

Observable road rules and consequences

Signals, limits, lane markings, crosswalks, and right-of-way are observable rules. The simulator records violations and physical consequences but leaves the behavior decision to the driver.

A collision disables the involved vehicle. The wreck remains a physical obstacle on its actual lane or connector; adjacent lanes can remain usable.

Wake events

`CurrentWake.new\_events` contains non-visual facts such as onboard radar warnings, command receipts, passenger requests and terminal state. Visual traffic facts are never delivered as semantic wake events. `active\_control` contains only ego commands that actually committed and remain relevant.

After `simulation\_ended`, newly submitted physical commands are discarded.

I.2 weather_transition

Skill: weather_transition

Weather Transition Rules

****Only act when you see a `[Weather]` event.**** Do not preemptively adjust settings -- the vehicle's defaults are correct for clear weather.

Required external equipment

Apply the entering-weather settings and any applicable cleanup below. Only operate installed equipment; an already satisfied state needs no repeat action. These instructions use the same rules as scoring and NPCs.

Skill: weather_transition

Entering `foggy`

- `lowBeamHeadlight.switch('on')`
- `highBeamHeadlight.switch(False)`
- `fogLight.carcontrol\_fogLight\_switch(True, 'front')`
- `fogLight.carcontrol\_fogLight\_switch(True, 'rear')`
- `positionLight.carcontrol\_positionLight\_switch(True)`

Entering `hail`

- `window.carcontrol\_window\_switch(['all'], False)`
- `sunroof.carcontrol\_sunroof\_switch('close')`

Entering `heavy\_rain`

- `wiper.carcontrol\_wiperBlade\_switch(True, 'front')`
- `window.carcontrol\_window\_switch(['all'], False)`
- `sunroof.carcontrol\_sunroof\_switch('close')`

Entering `heavy\_snow`

- `wiper.carcontrol\_wiperBlade\_switch(True, 'front')`
- `window.carcontrol\_window\_switch(['all'], False)`
- `sunroof.carcontrol\_sunroof\_switch('close')`
- `lowBeamHeadlight.switch('on')`
- `highBeamHeadlight.switch(False)`
- `fogLight.carcontrol\_fogLight\_switch(True, 'front')`
- `fogLight.carcontrol\_fogLight\_switch(True, 'rear')`
- `positionLight.carcontrol\_positionLight\_switch(True)`

Entering `rainy`

- `wiper.carcontrol\_wiperBlade\_switch(True, 'front')`
- `window.carcontrol\_window\_switch(['all'], False)`
- `sunroof.carcontrol\_sunroof\_switch('close')`

Entering `snowy`

- `wiper.carcontrol\_wiperBlade\_switch(True, 'front')`
- `window.carcontrol\_window\_switch(['all'], False)`
- `sunroof.carcontrol\_sunroof\_switch('close')`
- `lowBeamHeadlight.switch('on')`

Leaving `heavy\_rain`, `heavy\_snow`, `rainy`, `snowy` for a condition outside that set

- `wiper.carcontrol\_wiperBlade\_switch(False, 'front')`

Skill: weather_transition

Leaving `foggy`, `heavy\_snow` for a condition outside that set

- `fogLight.carcontrol\_fogLight\_switch(False, 'front')`
- `fogLight.carcontrol\_fogLight\_switch(False, 'rear')`
- `positionLight.carcontrol\_positionLight\_switch(False)`

During dusk/night/dawn, keep positionLight ON despite weather cleanup; the day/night lighting requirement takes precedence. Weather cleanup does not turn lowBeamHeadlight off. Weather cleanup does not reopen window or sunroof. Sunny/cloudy adds no entering-weather action; apply only relevant cleanup.

Cold weather (-> snowy/heavy_snow/hail)

- Turn ON steering wheel heater
- Turn ON rearview mirror heating
- Turn ON seat heater

Low visibility (-> foggy/heavy_rain/snowy/heavy_snow/hail)

- Enable HUD and reduce brightness

Leaving fog

- Turn OFF defrost and auto-defog

Leaving cold weather

- Turn OFF steering wheel heater, mirror heating, seat heater

Leaving low visibility

- Restore HUD brightness

I.3 daynight_transition

Skill: daynight_transition

Day/Night Transition Rules

****Only act when you see a `[DayNight]` event.**** Do not preemptively adjust settings based on the current time of day -- the vehicle's defaults are already correct for daytime.

Entering dark period (morning/noon/afternoon -> dusk/night/dawn)

****All four actions are required:****

1. `lowBeamHeadlight.switch('on')` -- headlights on
2. `positionLight.carcontrol\_positionLight\_switch(True)` -- position lights on
3. `centerInformationDisplay.brightness\_decrease(degree='large')` -- dim display
4. `rearviewMirror.mode\_autoAdjust(True)` -- anti-glare

Leaving dark period (dusk/night/dawn -> morning/noon/afternoon)

- Turn OFF low beam headlights: `lowBeamHeadlight.switch('off')`

Skill: daynight_transition

```
- Turn OFF position lights: `positionLight.carcontrol_positionLight_switch(False)`  
- Increase center display brightness: `centerInformationDisplay.brightness_increase(degree='large')`  
- Disable rearview mirror auto-adjust: `rearviewMirror.mode_autoAdjust(False)`  
  
## First tick in dark period  
- Same as entering dark period (treat as initial setup)
```

I.4 safety_refusal

Skill: safety_refusal

```
# Safety Refusal Rules  
  
## When to refuse  
Refuse and broadcast a safety warning when the passenger requests:  
- Opening doors while driving  
- Opening trunk while driving  
- Playing video while driving (distraction)  
- Any action that compromises vehicle safety  
  
## How to refuse  
- Do NOT execute the unsafe action  
- Call `broadcast.broadcast_safety_refusal(True, reason)` with a clear reason  
- Respond politely to the passenger explaining why the request cannot be fulfilled  
  
## HARD safety constraints (enforced by ConstraintEngine)  
These are blocked at the system level regardless:  
- Door open while speed > 0  
- Trunk open while speed > 0  
- Video playback while driving  
  
## Safe alternatives  
- If passenger wants to open door: suggest stopping first  
- If passenger wants video: offer audio-only alternatives (music, radio)
```

J Tool Inventory

J.1 Driving Agent (DA) Tools

DA tool schemas come from installed modules and the harness. The ten core calls are listed in Table 19; discovery and harness calls appear in Table 20. Other `module__method` calls are inspected and loaded on demand, with loaded schemas retained across wakes subject to the context budget.

Table 19 Core DA tools preloaded when their modules are installed.

Tool	Function
<code>navigation__navigation_route_plan</code>	Display a suggested route.
<code>navigation__navigation_minimap</code>	Show a route or local map without replanning.

Tool (continued)	Function
<code>navigation__navigation_set_speed</code>	Set target speed and optional acceleration limits.
<code>navigation__navigation_emergency_stop</code>	Request emergency braking.
<code>navigation__navigation_change_lane</code>	Request one adjacent lane.
<code>navigation__navigation_select_maneuver</code>	Select a legal junction connector.
<code>frontRadar__scan</code>	Read the front radar.
<code>rearRadar__scan</code>	Read the rear radar.
<code>speedLimit__speed_limit_get</code>	Read the current speed limit.
<code>turnSignal__switch</code>	Set a persistent turn signal.

DA discovers APIs for the installed subset of the modules listed in Table 15; absent equipment is omitted. The discovery catalog also includes `road_perception`, a virtual engine API. Its text tools omit camera-visible objects and signals; optional `lidar` supplies an image.

Table 20 DA discovery, harness, and additional tool functions.

Tool	Function
<code>get_module_api</code>	List a module’s callable methods.
<code>load_tools</code>	Load selected module APIs.
<code>load_skill</code>	Load procedural guidance.
<code>todo_manage</code>	Maintain tasks across wakes.
<code>memory_search</code>	Search passenger, event, and action history.
<code>set_heartbeat_interval</code>	Set periodic observation interval (0.1–30 s).
<code>schedule_next_wake</code>	Schedule a replaceable one-shot observation.
<code>get_device_state</code>	Read cabin equipment state.
<code>finish</code>	End the current DA wake.
<code>navigation__navigation_u_turn</code>	Request an optional U-turn.

Cabin calls report state changes; motion calls report acceptance and need later receipts or observation to confirm completion. Invalid calls return repairable errors.

J.2 Personal Agent (PA) Tools

PA’s four non-driving calls (Table 21) manage the passenger request sheet shared with DA and Judge.

Table 21 PA request-management tools.

Tool	Function
<code>send_immediate_request</code>	Create or replace an immediate request with outcomes and timing.
<code>send_triggered_request</code>	Create or replace a two-phase request with a typed trigger.
<code>cancel_passenger_request</code>	Cancel the active request when no pending outcome is wanted.
<code>finish</code>	End the wake; keep the active request unchanged.

J.3 Judge Tools

Judge’s submissions (Table 22) are validated against the frozen request and timeline; neither changes vehicle state or wakes DA.

Table 22 Judge submission tools.

Tool	Function
<code>submit_passenger_check</code>	Record intermediate criterion statuses and evidence; no grade.
<code>submit_passenger_judgement</code>	Submit a final A–F/NA grade, statuses, and fulfillment times.